

Wheat Blast on Wheat

Magnaporthe oryzae triticum



Wheat infected with the blast fungus (Feb 24, 2019, Meherpur, Bangladesh), © 2019 Kamoun et al., CC BY. <https://doi.org/10.1371/journal.pbio.3000302.g001>.

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
- It is seed transmitted so use disease free or certified seed - Use resistant varieties (Tyetye, Mpheta) - Adhere to recommended seeding rate of 100 kg/ha to avoid overcrowding - Destruction of alternative surrounding hosts (all of which are grasses) - Crop rotation with legumes or other non-grass crops will have some effect - It is most important to plant so as to avoid the rains during flowering or grain fill stages - Excessive levels of nitrogen fertilizers should be avoided to help reduce disease severity	- Begin inspection at seedling stage - Eye shape lesions with grey or whitish centres surrounded by dark brown margins on seedling leaves - Eye shaped or elliptical lesions on mature leaves and stems with white centres - Silvery bleached spikes with a green canopy - Dark-gray sporulation on the rachis leading to partial or full drying - Take action if the symptoms are seen, try to apply controls to the flag leaf (just as the ear is emerging).	Containment and quarantine in affected areas; do not move infected straw around.	- Alternate fungicides - When using a pesticide or botanical, always wear protective clothing and follow instructions on the product label such as dosage, timing of application, pre-harvest interval, maximum number of sprays - Do not eat or smoke while spraying the pesticides - It is important to protect the ear at flowering. So if the leaves are infected make sure sprays are on before the ear emerges. - Seed treatment with fungicides such as carboxin, difenoconazole - Prophylactic foliar spray of a combination of triazoles and strobilurins. for example epoxiconazole 50 g/l + pyraclostrobin 133 g/l e.g. Opera OR Radial 75g/l and 75g/l OPERA @0.5L/Ha OR RADIAL @ - Epoxiconazole (EPIC) 125 g/litre (12 % w/w) 1 L/Ha - Trifloxystrobin (250 g/kg) and Tebuconazole (500 g/kg) eg Nativo 75 WG use at 300g per Ha.	- Difenoconazole: WHO Class II (Moderately hazardous). Frac 3 : Carboxin Frac 7 WHO Class III - Epoxiconazole (DMI-fungicides) Frac 3 , PHI 28 days Pyraclostrobin (strobilurin) Frac 11.PHI 10 days - Epoxiconazole (DMI-fungicides) Frac 3 , PHI 28 days - Frac 3 and Frac 11 Max 2 applications per season



Zambia

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